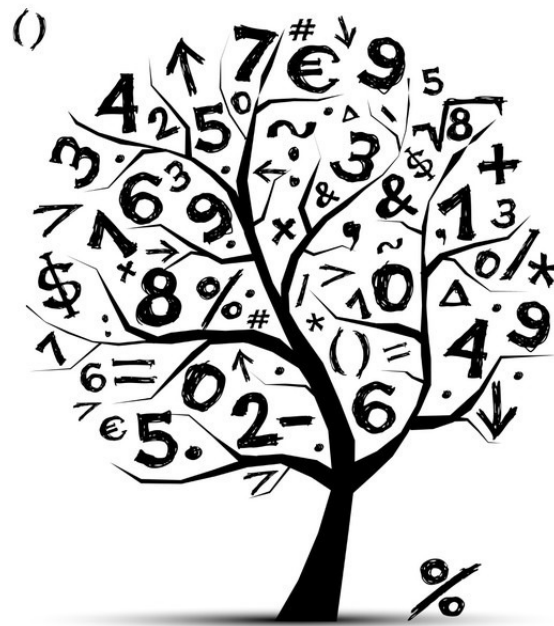


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Exercises: Matrices



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A square symmetric matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is positive semidefinite if:

$$\mathbf{x}' \mathbf{Q} \mathbf{x} \geq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

Exercise 1

Using the definition, prove that the matrix

$$\mathbf{Q}_1 = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

is positive semidefinite.

We have:

$$\begin{aligned} \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \left(\left(\frac{1}{3} x_1 + 0 \cdot x_2 \right) \quad \left(0 \cdot x_1 + \frac{1}{2} x_2 \right) \right) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= \left(\frac{1}{3} x_1 \right) x_1 + \left(\frac{1}{2} x_2 \right) x_2 \\ &= \frac{1}{3} x_1^2 + \frac{1}{2} x_2^2 \geq 0, \quad \forall (x_1, x_2) \in \mathbb{R}^2 \end{aligned}$$

Accordingly, the matrix $\mathbf{Q}_1$ is positive semidefinite.

Exercise 2

Using the definition, prove that the matrix

$$\mathbf{Q}_2 = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

is positive semidefinite.

We have:

$$\begin{aligned}
 (x_1 \quad x_2) \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= ((3x_1 + x_2) \quad (x_1 + 3x_2)) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\
 &= (3x_1 + x_2)x_1 + (x_1 + 3x_2)x_2 \\
 &= 3x_1^2 + x_1x_2 + x_1x_2 + 3x_2^2 \\
 &= 3x_1^2 + 2x_1x_2 + 3x_2^2 \\
 &= (x_1 + x_2)^2 + 2x_1^2 + 2x_2^2 \geq 0, \quad \forall (x_1, x_2) \in \mathbb{R}^2
 \end{aligned}$$

accordingly the matrix $\mathbf{Q}_2$ is positive semidefinite.

Exercise 3

Using the definition, prove that the matrix

$$\mathbf{Q}_3 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is positive semidefinite.

We have:

$$\begin{aligned}
 & (x_1 \ x_2 \ x_3) \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\
 &= \begin{pmatrix} (2x_1 - x_2) & (-x_1 + 2x_2 - x_3) & (-x_2 + 2x_3) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\
 &= (2x_1 - x_2)x_1 + (-x_1 + 2x_2 - x_3)x_2 + (-x_2 + 2x_3)x_3 \\
 &= 2x_1^2 - x_2x_1 - x_1x_2 + 2x_2^2 - x_3x_2 - x_2x_3 + 2x_3^2 \\
 &= 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2 \\
 &= x_1^2 + x_1^2 - 2x_1x_2 + x_2^2 + x_2^2 - 2x_2x_3 + x_3^2 + x_3^2 \\
 &= x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_3^2 \geq 0, \quad \forall (x_1, x_2, x_3) \in \mathbb{R}^3
 \end{aligned}$$

accordingly the matrix $\mathbf{Q}_3$ is positive semidefinite.

Exercise 4

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 1 & 0 \\ -3 & 2 & 5 \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

Compute $\mathbf{A} + \mathbf{B}$, $2\mathbf{A} - \mathbf{B}$, $\mathbf{A}'$, and verify that $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$.

Sum and scalar multiplication are computed entry by entry:

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 1+2 & -2+1 & 3+0 \\ 0-3 & 4+2 & -1+5 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 3 \\ -3 & 6 & 4 \end{pmatrix},$$

$$2\mathbf{A} - \mathbf{B} = \begin{pmatrix} 2 & -4 & 6 \\ 0 & 8 & -2 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 0 \\ -3 & 2 & 5 \end{pmatrix} = \begin{pmatrix} 0 & -5 & 6 \\ 3 & 6 & -7 \end{pmatrix}.$$

The transpose is obtained by interchanging rows and columns ($\mathbf{A}' \in \mathbb{R}^{3 \times 2}$):

$$\mathbf{A}' = \begin{pmatrix} 1 & 0 \\ -2 & 4 \\ 3 & -1 \end{pmatrix}, \quad \mathbf{B}' = \begin{pmatrix} 2 & -3 \\ 1 & 2 \\ 0 & 5 \end{pmatrix}.$$

Finally,

$$\mathbf{A}' + \mathbf{B}' = \begin{pmatrix} 3 & -3 \\ -1 & 6 \\ 3 & 4 \end{pmatrix} = (\mathbf{A} + \mathbf{B})'.$$

Exercise 5

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{pmatrix} \in \mathbb{R}^{2 \times 3}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ -1 & 3 \end{pmatrix} \in \mathbb{R}^{3 \times 2}, \quad \mathbf{C} = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \in \mathbb{R}^{2 \times 2}.$$

1. Among the products $\mathbf{AB}$, $\mathbf{BA}$, $\mathbf{AC}$, $\mathbf{CA}$, $\mathbf{BC}$, $\mathbf{CB}$, say which ones are defined and give the dimensions of the result.
2. Compute $\mathbf{CA}$, $\mathbf{AB}$ and $\mathbf{BC}$.

1. The product of a matrix in $\mathbb{R}^{m \times n}$ and a matrix in $\mathbb{R}^{q \times p}$ is defined only if $n = q$ (number of columns of the first = number of rows of the second), and the result is in $\mathbb{R}^{m \times p}$. Hence:

- **AB** : $(2 \times 3)(3 \times 2)$, defined, result in $\mathbb{R}^{2 \times 2}$;
- **BA** : $(3 \times 2)(2 \times 3)$, defined, result in $\mathbb{R}^{3 \times 3}$;
- **AC** : $(2 \times 3)(2 \times 2)$, **not** defined ($3 \neq 2$);
- **CA** : $(2 \times 2)(2 \times 3)$, defined, result in $\mathbb{R}^{2 \times 3}$;
- **BC** : $(3 \times 2)(2 \times 2)$, defined, result in $\mathbb{R}^{3 \times 2}$;
- **CB** : $(2 \times 2)(3 \times 2)$, **not** defined ($2 \neq 3$).

2. Each entry (i, k) is the scalar product of row i of the first matrix and column k of the second one:

$$CA = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 3 & 1 \cdot 0 + 2 \cdot (-1) & 1 \cdot 2 + 2 \cdot 1 \\ 0 \cdot 1 + (-1) \cdot 3 & 0 \cdot 0 + (-1) \cdot (-1) & 0 \cdot 2 + (-1) \cdot 1 \end{pmatrix} = \begin{pmatrix} 7 & -2 & 4 \\ -3 & 1 & -1 \end{pmatrix},$$

$$AB = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 2 + 2 \cdot (-1) & 1 \cdot 0 + 0 \cdot 1 + 2 \cdot 3 \\ 3 \cdot 1 + (-1) \cdot 2 + 1 \cdot (-1) & 3 \cdot 0 + (-1) \cdot 1 + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} -1 & 6 \\ 0 & 2 \end{pmatrix},$$

$$BC = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 2 + 0 \cdot (-1) \\ 2 \cdot 1 + 1 \cdot 0 & 2 \cdot 2 + 1 \cdot (-1) \\ -1 \cdot 1 + 3 \cdot 0 & -1 \cdot 2 + 3 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 3 \\ -1 & -5 \end{pmatrix}.$$

Exercise 6

1. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Compute **AB** and **BA** . Is the product commutative?

2. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Compute **AB** . What do you observe?

1.

$$\mathbf{AB} = \begin{pmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 0 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 1 + 0 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Since $\mathbf{AB} \neq \mathbf{BA}$, the product is not commutative (even for square matrices of the same size).

2.

$$\mathbf{AB} = \begin{pmatrix} 1 \cdot 1 + 1 \cdot (-1) & 1 \cdot (-1) + 1 \cdot 1 \\ 1 \cdot 1 + 1 \cdot (-1) & 1 \cdot (-1) + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The product of two non-zero matrices can be the zero matrix: unlike real numbers, $\mathbf{AB} = \mathbf{0}$ does **not** imply $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$.

Exercise 7

1. For each of the following matrices, say whether it is square, diagonal, symmetric, upper triangular, lower triangular:

$$\mathbf{M}_1 = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}, \quad \mathbf{M}_2 = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \mathbf{M}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ -1 & 3 & 4 \end{pmatrix}, \quad \mathbf{M}_4 = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

2. Compute the product of the upper triangular matrices $\mathbf{U}_1 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $\mathbf{U}_2 = \begin{pmatrix} 2 & -1 \\ 0 & 4 \end{pmatrix}$. What kind of matrix do you obtain?
3. Let $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Compute $\mathbf{DA}$ and $\mathbf{AD}$ and describe the effect of the diagonal matrix.

1.
 - M_1 : square and symmetric ($m_{12} = m_{21} = 2$); it is neither diagonal nor triangular.
 - M_2 : square and diagonal; hence it is also symmetric, upper triangular and lower triangular.
 - M_3 : square and lower triangular (all entries above the main diagonal are zero); it is not symmetric (e.g., $m_{21} = 5 \neq 0 = m_{12}$).
 - $M_4 \in \mathbb{R}^{2 \times 3}$: not square, hence none of the other properties applies.

2.

$$U_1 U_2 = \begin{pmatrix} 1 \cdot 2 + 2 \cdot 0 & 1 \cdot (-1) + 2 \cdot 4 \\ 0 \cdot 2 + 3 \cdot 0 & 0 \cdot (-1) + 3 \cdot 4 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 0 & 12 \end{pmatrix},$$

which is again upper triangular, with diagonal entries $1 \cdot 2 = 2$ and $3 \cdot 4 = 12$.

3.

$$DA = \begin{pmatrix} 2 \cdot 1 & 2 \cdot 2 \\ 3 \cdot 3 & 3 \cdot 4 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 9 & 12 \end{pmatrix}, \quad AD = \begin{pmatrix} 1 \cdot 2 & 2 \cdot 3 \\ 3 \cdot 2 & 4 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 6 & 12 \end{pmatrix}.$$

Left multiplication by D multiplies the i -th **row** of A by d_{ii} ; right multiplication multiplies the j -th **column** of A by d_{jj} .

Exercise 8

Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

and let $\pi = (2, 3, 1)$.

1. Write the permutation matrix P_π and compute $P_\pi A$.
2. Compute AP'_π and check that the columns of A are reordered according to π .
3. Verify that $P_\pi P'_\pi = I_3$.

1. P_π is obtained from I_3 by placing row 2 first, then row 3, then row 1:

$$P_\pi = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad P_\pi A = \begin{pmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{pmatrix}.$$

The rows of A appear in the order 2, 3, 1.

- 2.

$$AP'_\pi = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \\ 8 & 9 & 7 \end{pmatrix}.$$

The columns of A appear in the order 2, 3, 1.

- 3.

$$P_\pi P'_\pi = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

Hence P_π is invertible and $P_\pi^{-1} = P'_\pi$ (one checks in the same way that $P'_\pi P_\pi = I_3$).

Exercise 9

Compute the determinant of

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & -2 \\ 1 & 4 & 5 \end{pmatrix}$$

using the Laplace expansion along the most convenient row or column, and check the result with the Sarrus rule.

The second column contains two zeros, so we expand along it: only the entry $a_{32} = 4$ contributes,

$$\det(A) = 0 \cdot C_{12} + 0 \cdot C_{22} + 4 \cdot C_{32}.$$

The cofactor is

$$C_{32} = (-1)^{3+2} \det \begin{pmatrix} 2 & 1 \\ 3 & -2 \end{pmatrix} = -(2 \cdot (-2) - 1 \cdot 3) = -(-4 - 3) = 7.$$

Therefore $\det(A) = 4 \cdot 7 = 28$.

Check with Sarrus ($a = 2, b = 0, c = 1, d = 3, e = 0, f = -2, g = 1, h = 4, i = 5$):

$$\det(A) = 2 \cdot 0 \cdot 5 + 0 \cdot (-2) \cdot 1 + 1 \cdot 3 \cdot 4 - 1 \cdot 0 \cdot 1 - 0 \cdot 3 \cdot 5 - 2 \cdot (-2) \cdot 4 = 0 + 0 + 12 - 0 - 0 + 16 = 28.$$

Exercise 10

Compute the determinant of

$$A = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 2 & 0 \\ 4 & 1 & 0 & 2 \\ 1 & 0 & 1 & 3 \end{pmatrix} \in \mathbb{R}^{4 \times 4}$$

choosing the row or column that minimizes the number of computations.

The second row has only one non-zero entry, $a_{23} = 2$. Expanding along the second row:

$$\det(A) = a_{23} C_{23} = 2 \cdot (-1)^{2+3} \det(A_{23}) = -2 \det(A_{23}),$$

where A_{23} is obtained by deleting row 2 and column 3:

$$A_{23} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 1 & 2 \\ 1 & 0 & 3 \end{pmatrix}.$$

We expand $\det(A_{23})$ along its third row, which contains a zero:

$$\begin{aligned} \det(A_{23}) &= 1 \cdot (-1)^{3+1} \det \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} + 0 + 3 \cdot (-1)^{3+3} \det \begin{pmatrix} 1 & 2 \\ 4 & 1 \end{pmatrix} \\ &= 1 \cdot (4 - 3) + 3 \cdot (1 - 8) = 1 - 21 = -20. \end{aligned}$$

Therefore

$$\det(A) = -2 \cdot (-20) = 40.$$

Exercise 11

Consider the matrices

$$U = \begin{pmatrix} 2 & 5 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 7 & 2 & 0 \\ -3 & 4 & 5 \end{pmatrix}.$$

1. Compute $\det(U)$ and $\det(L)$.
2. Without computing any product, find $\det(UL)$, $\det(U')$, $\det(3U)$ and $\det(U^{-1})$. Why does U^{-1} exist?
3. Verify that $\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$ is the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$, and check that $\det(A^{-1}) = 1/\det(A)$.

1. Both matrices are triangular, so their determinant is the product of the diagonal entries:

$$\det(\mathbf{U}) = 2 \cdot (-3) \cdot 1 = -6, \quad \det(\mathbf{L}) = 1 \cdot 2 \cdot 5 = 10.$$

2. Using the properties of determinants (with $n = 3$):

$$\det(\mathbf{UL}) = \det(\mathbf{U}) \det(\mathbf{L}) = -6 \cdot 10 = -60, \quad \det(\mathbf{U}') = \det(\mathbf{U}) = -6,$$

$$\det(3\mathbf{U}) = 3^3 \det(\mathbf{U}) = 27 \cdot (-6) = -162.$$

Since $\det(\mathbf{U}) = -6 \neq 0$, the matrix $\mathbf{U}$ is invertible, and $\det(\mathbf{U}^{-1}) = \frac{1}{\det(\mathbf{U})} = -\frac{1}{6}$.

- 3.

$$\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 6-5 & -3+3 \\ 10-10 & -5+6 \end{pmatrix} = \mathbf{I}_2,$$

$$\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & 2-2 \\ -15+15 & -5+6 \end{pmatrix} = \mathbf{I}_2.$$

Hence it is the inverse of $\mathbf{A}$. Moreover $\det(\mathbf{A}) = 3 \cdot 2 - 1 \cdot 5 = 1$ and $\det(\mathbf{A}^{-1}) = 2 \cdot 3 - (-1) \cdot (-5) = 1 = \frac{1}{1}$.

Exercise 12

Compute the rank of the following matrices:

$$\mathbf{A}_1 = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}, \quad \mathbf{A}_2 = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{pmatrix}, \quad \mathbf{A}_3 = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

- A_1 : the second column is twice the first one, $(2 \ 6)' = 2(1 \ 3)'$, so the two columns are linearly dependent; the first column is non-zero, hence it is linearly independent by itself. Therefore $\text{rank}(A_1) = 1$ (consistently, $\det(A_1) = 6 - 6 = 0$).
- A_2 : the third column is the sum of the first two, $(3 \ 1 \ 4)' = (1 \ 0 \ 1)' + (2 \ 1 \ 3)'$, so the three columns are linearly dependent and $\text{rank}(A_2) \leq 2$. The first two columns are linearly independent: $\lambda_1(1 \ 0 \ 1)' + \lambda_2(2 \ 1 \ 3)' = \mathbf{0}$ gives $\lambda_2 = 0$ (second entry) and then $\lambda_1 = 0$ (first entry). Hence $\text{rank}(A_2) = 2$. Consistently, expanding along the first row, $\det(A_2) = 1 \cdot (4 - 3) - 2 \cdot (0 - 1) + 3 \cdot (0 - 1) = 1 + 2 - 3 = 0$.
- A_3 : expanding along the first row,

$$\det(A_3) = 1 \cdot (1 \cdot 0 - 1 \cdot 1) - 0 \cdot (0 \cdot 0 - 1 \cdot 1) + 2 \cdot (0 \cdot 1 - 1 \cdot 1) = -1 - 0 - 2 = -3 \neq 0.$$

Since A_3 is square and $\det(A_3) \neq 0$, we have $\text{rank}(A_3) = 3$.